

SUPPLY CHAIN MANAGEMENT

Introduction and objectives of supply chain, Decision phases in a supply chain, Purchasing tools and techniques, Value analysis, Project planning and control techniques, Pricing and revenue management, Costing fundamentals, Types of costing, Managing inventory in a supply chain, Economic order quantity, EOQ determination with instantaneous delivery and without shortages, Effect of quantity discount, safety stock, reorder level & lead time. Facility decisions in supply chain: Factors influencing network design in supply chain, Models for facility location and capacity allocation, Transportation decisions in a supply chain, Routing and scheduling in transportation, Multistage transportation problems, Truck allocation problem, Travelling salesman problem, Vehicle routing problems, Financial evaluation of supply chain Decisions, The impact of financial factors on supply chain decisions, Discounted cash flow analysis, Evaluating supply chain decisions using decision trees.

SOCIAL NETWORK ANALYSIS

Introduction to Social Web, Nodes, Edges and Network Measures, Describing Nodes and Edges, Describing Networks, Layouts, Visualizing network features, The role of Tie strength, Measuring Tie strength and its network structures, network propagation, Link prediction, entity resolution, Case study, Introduction to community discovery, communities in context, quality functions, The Kernighan-Lin algorithm, Agglomerative algorithms, spectral algorithms, multi-level graph partitioning, Markov clustering, Other approaches, Introduction to social influence, Influence related statistics, social similarity and influence, Homophily, Existential Test for social influence, Influence and actions, Influence and interactions, influence maximization in viral marketing.

INFORMATION RETRIEVAL

Introduction to Information Retrieval: Mathematical Basics, Vector spaces and Similarity, Probabilities and Statistics, Text Analysis; Pre-processing: Document processing, Stemming, String Matching, Basic NLP tasks – POS Tagging Shallow Parsing; Overview of Text Retrieval Systems: System Architecture, Boolean Models, Inverted Indexes, Document Ranking, IR Evaluation; Retrieval Models and Implementation: Vector Space Models, TF-IDF Weighting, Retrieval Axioms, Implementation Issues, Probabilistic Models; Statistical Language Models: Okapi/BM25, Language Models, KL-divergence, Smoothing; Query Expansion and Feedback: Query Reformulation, Relevance feedback, PseudoRelevance Feedback, Language Model Based, Feedback; Web Search Engines: Models of the Web, Web Crawling; Static Ranking: Page Rank HITS, Query Log Analysis, Adversarial IR, Information Filtering: Adaptive Filtering, Collaborative Filtering, User Interfaces, Text Classification, Naïve Bayes, K-nearest neighbors, Feature selection, Semi-supervised Learning; Text Clustering: Vector-space Clustering; K-means, EM algorithm, Text shingling; Graph-Based Methods: WordNet, Document and Word Graphs, Network Analysis, Random Walks, Harmonic Functions.

BLOCKCHAIN TECHNOLOGY

Structure of a Block, The Genesis Block, Linking Blocks in the Blockchain, Merkle Trees. Simplified Payment Verification: Using hash functions to chain blocks, for Proof-of-Work, Digital Signatures to sign transactions, Distributed Ledger, Byzantine Agreement, Eventual Consistency & Bitcoin Consistency- Availability and Partitions, Bitcoin, Smart Contracts, Weak Consistency, Distributed Storage. Consistent Hashing: Hypercubic Networks, Mining and Consensus: Decentralized Consensus, Independent Verification of Transactions Mining Nodes, Aggregating Transactions into Blocks, Constructing the Block Header, Successfully Mining the Block, Evaluating a New Block, Assembling and Selecting Chains of Blocks. Consensus Attacks: Changing the Consensus Rules, Soft Fork Signaling with Block Version, Consensus Software Development, Ethereum and Bitcoin, block format, mining algorithm, proof-of-stake (PoS) algorithm, account management, contracts and transactions, Solidity language. Account management: contracts and transactions, Applications of Blockchain: Case studies.

COMPUTER VISION

Introduction : Image Processing, Components of Image processing system, Image formation and digitization concepts, Neighbours of pixel adjacency connectivity, regions and boundaries, Distance measures, Image processing operations, Arithmetic, Logical, Geometrical, Convolution and Correlation Operations, Applications: Document Image Analysis, Biometrics, Object Recognition, Tracking, Medical Image Analysis, Content-Based Image Retrieval, Video Data Processing, Multimedia, Virtual Reality and Augmented Reality, Image Formation, Image representations (continuous and discrete) , Image pre-processing Techniques, Feature Extraction-Point, Line and Edge Detection, Color, Texture, Shape and structure Features in spatial and frequency domains, Corner Detection, Hough Transform , Image Segmentation: Boundary detection based techniques, Point, line detection, Edge detection, Edge linking, local processing, regional processing, Hough transform, Thresholding, Iterative thresholding, Otsu's method, Moving averages, Multivariable thresholding, Region-based segmentation, Watershed algorithm, Use of motion in segmentation, Computer Vision: Computer Vision, What is Computer Vision - Low-level, Mid-level, High-level , Overview of Diverse Computer Vision Applications, Fundamentals of object recognition, Low-level computer vision-Edges, contours, textures, shapes, and colors , Motion, optical flow, and tracking Local features, invariance, bag-of-words models, Fisher vector, Middle-level representations of objects: parts, attributes, embedding.

SOFTWARE ENGINEERING

Introduction to Software Engineering: Software Components, Software Characteristics, Software Crisis, Software Engineering Processes; Software Development Life Cycle (SDLC) Models: Water Fall Model, Prototype Model, Spiral Model; Requirement Engineering Process: Analysis, Documentation, Data Flow Diagrams, SRS Document, IEEE Standards for SRS; Estimation: Estimation of Various Parameters Such as Cost, Efforts, Use Case Point, Class Point Method Estimating, Constructive Cost Models (COCOMO) and its Types, Function Point, Reusability, Object Points, The Putnam Resource Allocation Model; Software Design: Architectural Design, Low Level Design, Modularization, Design Structure Charts, Pseudo Codes, Coupling and Cohesion Measures, Function Oriented Design, Object Oriented Design; Coding Standards: Code Review and Walkthrough, Code Inspection; Testing : Testing Objectives, Testing and Debugging, Various Types of Testing Methods; Software Maintenance: Categories of Maintenance, Maintenance Process, Maintenance Models, Estimation of Maintenance Cost, Regression Testing, Software Re-Engineering, Reverse Engineering, Configuration Management and Certification; Software Quality: The Management Spectrum- (The People, The Product, The Process, The Project), SEI Capability Maturity Model, Verification and Validation, SQA Plans, Software Quality Frameworks, ISO 9126 Model, Certification Case Study.